Schema-Aware Grounding Effects in PostGIS Natural-Language-to-SQL: A Subset-Decomposed Evaluation Across Eleven LLMs

Ning Zhou
Beijing SuperMap Software Co., Ltd., Beijing, China
ORCID: 0009-0002-5647-7388
Corresponding author: zhouning1@supermap.com

Abstract

Natural-language interfaces to spatial databases increasingly use schema-aware grounding: prompt-side evidence about table aliases, geometry columns, spatial reference systems, PostGIS operators, value hints, and safety constraints. For computational geoscience applications, this is a spatial-database informatics problem rather than only a language-model benchmarking problem. We evaluate grounding as an intervention on CQ-125, a 125-question Chongqing PostGIS benchmark containing 85 spatial-query questions and 40 robustness questions, across 11 LLM families with N=3 paired samples per family. All paired tests use question-level majority vote across the stochastic samples followed by a single exact two-sided McNemar test, avoiding the pseudoreplication that results from treating NQ repeated samples as independent observations. Grounding produces a subset-dependent pattern. On the Robustness subset, all 11 families improve in point estimate (+12.50 to +65.00 pp; nine significant in unadjusted per-cell tests and after Holm correction within the Robustness comparison set, one marginal, one non-significant positive). On the Spatial subset, 7 families show unadjusted significant positive gains (+11.76 to +23.53 pp), 2 show marginal positive gains, and 2 show non-significant negative point estimates; Holm correction within the Spatial comparison set leaves 3 positive cells significant. The most negative panel estimate is `gemini-3.5-flash` ($\Delta = -10.59$ pp, $p = 0.136$), and a balanced N=5 focused check on the same family retains this negative but non-significant Spatial point estimate. We therefore make a bounded claim: subset decomposition and de-pooled paired testing are necessary for defensible grounding-effect evaluation in geospatial NL2SQL; cross-family effects are heterogeneous; and the observed negative tail requires out-of-benchmark replication before it can be treated as a stable model-family property.

Keywords: NL2GeoSQL; PostGIS; semantic grounding; spatial SQL; LLM evaluation; reproducibility.

1 Introduction

Natural language access to spatial databases is a long-standing GIScience problem (Goodchild, 1992; Egenhofer and Franzosa, 1991; Clementini et al., 1993). Large language models have advanced general text-to-SQL on Spider (Yu et al., 2018) and BIRD (Li et al., 2023), but PostGIS NL2GeoSQL also requires OGC Simple Feature Access predicates (Open Geospatial Consortium, 2010), coordinate-reference transformations, the `geometry`/`geography` distinction, KNN operators, and production safety constraints (PostGIS Development Team, 2024). A correct system must therefore translate geographic concepts into executable spatial predicates while preserving units, coordinate-reference assumptions, result shape, and read-only behaviour.

The standard engineering response is *schema-aware grounding*: an upstream stage that injects table aliases, value enumerations, geometry-type metadata, SRID flags, large-table boundedness rules, and PostGIS dialect cues into the LLM prompt. Grounding is often evaluated as a single positive intervention, but this aggregate view is too coarse for mixed geospatial benchmarks. A benchmark that combines ordinary spatial queries with robustness cases is not testing one homogeneous capability. Spatial questions reward precise predicate choice, join structure, geometry measurement, and output-shape control; robustness questions reward refusal, schema enforcement, and bounded execution. The same grounding context can therefore interact differently with each subset.

We evaluate that interaction on CQ-125, a 125-question Chongqing PostGIS benchmark with 85 Spatial and 40 Robustness questions. Across 11 LLM families at N=3 paired samples per family, grounding has positive Robustness point estimates for every family (+12.50 to +65.00 pp). Spatial effects are heterogeneous: seven families improve in unadjusted per-cell tests, two are marginal positive, and two have non-significant negative point estimates; after Holm correction within the 11 Spatial tests, three positive cells remain significant. The most negative panel estimate is `gemini-3.5-flash` (−10.59 pp, $p = 0.136$). A balanced N=5 three-condition check on this family retains the same Spatial grounding-only point estimate and a large Robustness gain (+27.50 pp, $p = 0.001$). Thus the claim is not that grounding generally hurts, but that grounding effects are subset-conditional and family-conditional.

Contributions.

1. **Methodological (§6.6).** Aggregate execution accuracy is insufficient for grounding-effect claims on geospatial NL2SQL when robustness and spatial questions are mixed. The paired-test convention also

matters: pooling N stochastic samples $\times$ Q questions into a single McNemar test on NQ observations violates independence and inflates significance. We use question-level majority vote across N samples followed by a single exact two-sided McNemar test on the per-question table.

2. **Empirical (§6).** On 11 LLM families surveyed at $N{=}3$ on CQ-125, grounding improves Robustness in all families by point estimate, but Spatial effects range from -10.59 to $+23.53$ pp. This establishes cross-family heterogeneity rather than a universal positive or negative effect.

3. **Diagnostic and design implication (§6.4–§6.5).** For `gemini-3.5-flash`, a focused three-condition analysis identifies a candidate negative-tail case and links many regressions to unsolicited SQL elaboration. A perfect-classifier routing calculation gives an upper bound for subset-conditional grounding, but we treat it as a design implication, not a validated production policy.

The semantic-layer framework, intent router, SQL postprocessor, and execution-time self-correction loop are described in §3. The central questions are *when grounding helps which question class on which family* and *how to test that effect with a valid paired statistic*. As supporting deployment context, §5.2 reports a fixed-host Gemma4 model-size sweep.

The paper next reviews related work, summarises the framework, describes the experiments, reports supporting framework results, presents the subset-decomposed grounding analysis, and discusses limitations and implications.

2 Related Work

2.1 GIScience foundations for NL-to-spatial-database

Natural language access to spatial databases sits within the GIScience programme articulated by Goodchild (1992): formalising geographic concepts so they can be computed rigorously. The relational semantics we ground against trace to the 4- and 9-intersection formalisms (Egenhofer and Franzosa, 1991), the end-user topological relations of Clementini et al. (1993), OGC Simple Feature Access (Open Geospatial Consortium, 2010), and GeoSPARQL (Open Geospatial Consortium, 2012). PostGIS (PostGIS Development Team, 2024) operationalises these ideas in PostgreSQL through geometry/geography types, casts, metric and topological functions, coordinate transformations, and KNN operators. Our grounding rules expose these constraints as prompt-side evidence.

Table 1: Direct comparison with recent NL-to-spatial-database systems. NALSpatial is implemented over SECONDO, not PostGIS. ✓: supported / reported; −: not reported.

System	E2E	Function	Safety	Paired gnd.	Released
Our framework	✓	−	✓	✓	✓
NALSpatial (Liu et al., 2025b)	−	−	−	−	✓
Monkuu (Yu et al., 2025)	✓	−	−	−	−
GeoSQL-Eval (Hou et al., 2026)	−	✓	−	−	✓
GeoCogent (Hou et al., 2025)	−	−	−	−	−
GeoAgent (Lin et al., 2026)	−	−	−	−	−

2.2 Natural language interfaces to spatial databases

Early geospatial question answering focused on small fact databases: Zelle and Mooney (1996) used inductive logic programming for GeoQuery, and Punjani et al. (2018) extended this to template-based question answering over linked geospatial data, highlighting spatial-predicate coverage and coordinate-system awareness. Recent LLM-era work falls into two groups. The first targets end-to-end NL-to-executable-spatial-SQL: NALSpatial (Liu et al., 2025b) translates user queries into spatial SQL (implemented over the SECONDO spatial DBMS rather than PostGIS); Monkuu (Yu et al., 2025) is an IJGIS-published LLM-powered NL interface with dynamic schema mapping; GeoSQL-Eval (Hou et al., 2026) contributes a large-scale LLM evaluation on PostGIS functions and NL2GeoSQL tasks. The second targets higher-level GeoAI agents: GeoCogent (Hou et al., 2025) is an LLM-based agent for geospatial code generation; GeoAgent (Lin et al., 2026) is a hierarchical multi-agent architecture. Surveys map the broader opportunities (Mai et al., 2024).

Our contribution is narrower than a general GeoAI agent and different from a function-coverage benchmark. We study grounding as an intervention in end-to-end execution-graded NL-to-PostGIS-SQL, where the semantic layer encodes OGC predicates, SRIDs, geography casts, KNN operators, and read-only bounded-output constraints. Compared with NALSpatial and Monkuu, we add a 125-question execution benchmark with an explicit Robustness suite, question-level McNemar testing, and an 11-family grounding-effect panel. Compared with GeoSQL-Eval, CQ-125 is smaller but designed for paired intervention analysis rather than broad function inventory.

Table 1 summarises the distinction.

2.3 Text-to-SQL with LLMs and semantic layers

LLM text-to-SQL has progressed through Spider (Yu et al., 2018; Rajkumar et al., 2022), BIRD (Li et al., 2023), Spider 2.0 (Lei et al., 2025), and BEAVER (Chen et al., 2024). DIN-SQL (Pourreza and Rafiei, 2023) decomposes prompting into schema linking, classification, generation, and self-correction; DAIL-SQL (Gao et al., 2024) uses skeleton-similarity few-shot selection; C3 (Dong et al., 2023) studies calibrated zero-shot prompting; CSpider (Min et al., 2019) is an early multilingual benchmark. Semantic-layer tools such as MetricFlow (dbt Labs, 2026) encode entities, measures, dimensions, and metric graphs for analytical warehouses. We adapt that vocabulary with GIS annotations, and use BIRD as a supporting warehouse track of the bilingual framework.

3 Method: Semantic-Layer Framework

3.1 Formal scope: semantic layer as metadata graph $G = (V, E)$

We formalise the semantic layer as a directed multigraph $G = (V, E)$ that encodes the executable subset of OGC Simple Feature Access (Open Geospatial Consortium, 2010) required for PostGIS grounding. Vertices cover geographic entities (tables with geometry column, SRID, and geometry type), warehouse dimensions/measures, intent tags, OGC predicate classes, and constraints for units, SRID ranges, safety, and refusal. Edges encode table–geometry links, foreign keys, predicate–entity links, intent routing, safety rules, and SRID-to-cast-to-unit chains. Algorithm 1 materialises G at startup from the live PostGIS catalog. G is an *executable subset* of the OGC SFA predicate taxonomy, not a complete geospatial ontology; Table 2 documents the enforced rule-level mapping.

3.2 Problem formulation

We formulate cross-domain natural language to SQL as translating a user question q into executable SQL s over either geospatial or warehouse databases. Let S be the catalog-derived schema representation and E be semantic evidence such as aliases, hierarchy expansions, geometry metadata, value hints, or fact/dimension/entity annotations:

$$\hat{s} = f(q, S, E), \tag{1}$$

Algorithm 1 BUILDSEMANTICGRAPH($schema, prefix$)

$G \leftarrow$ empty multigraph
for $(t, c, srid, \tau) \in$ `geometry_columns` **do**
  if t starts with $prefix$ and $srid \neq 0$ **then**
    add GEO_ENTITY vertex $(t, c, srid, \tau)$
  end if
end for
for $(t_s, t_t) \in$ `key_column_usage` **do**
  add edge $(t_s \rightarrow t_t, \mathrm{FK})$
end for
register topological / metric / KNN predicate vertices (OGC SFA 06-104r4)
register intent vertices; route intents $\rightarrow$ predicate classes
register unit-rule edges: metric $\rightarrow$ SRID range $\rightarrow$ required cast
return G

Table 2: Alignment of semantic-layer rules to OGC SFA (Open Geospatial Consortium, 2010), ISO 19107 (ISO/TC 211, 2019) clauses, and PostGIS functions (PostGIS Development Team, 2024). Rule-level mapping; each enforced rule traces to a specific clause.

Rule	Edge class	OGC 06-104r4 clause	ISO 19107 clause	PostGIS
`R_topo_intersects`	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	`ST_Intersects`
`R_topo_contains`	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	`ST_Contains`
`R_topo_within`	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	`ST_Within`
`R_metric_length_geog`	E_{unit}	§6.1.2.3 (Length) + §8.2.3	§6.4.14	`ST_Length(geog)`
`R_metric_area_geog`	E_{unit}	§6.1.2.3 (Area) + §8.2.3	§6.4.15	`ST_Area(geog)`
`R_metric_dwithin_geog`	E_{unit}	§6.1.2.3 (Distance) + §8.2.3	§6.4.10	`ST_DWithin(geog)`
`R_knn_nearest`	E_{knn}	§6.1.2.3 (Distance)	§6.4.10	<-> KNN
`R_safety_refusal`	E_{safety}	not in OGC	—	postprocessor AST
`R_safety_bounded`	E_{safety}	not in OGC	—	`LIMIT` injection

where f is an LLM-conditioned generation pipeline. Warehouse queries mainly require schema linking, joins, aggregation, and temporal filtering; geospatial queries also require geometry columns, spatial relations, coordinate systems, and operators for intersection, containment, buffering, distance, and area. The framework therefore mediates grounding by domain rather than using one homogeneous schema prompt.

3.3 Three-stage pipeline

NL2Semantic2SQL has three stages (Figure 1). Stage 1 resolves table and column annotations, aliases, domain hints, and task metadata. Stage 2 combines this evidence with schema inspection and optional few-shot retrieval to build the grounded prompt. Stage 3 postprocesses, safely executes, and, if needed, revises SQL through execution feedback. The decomposition separates semantic interpretation from SQL synthesis while keeping generation tied to executable schema information.

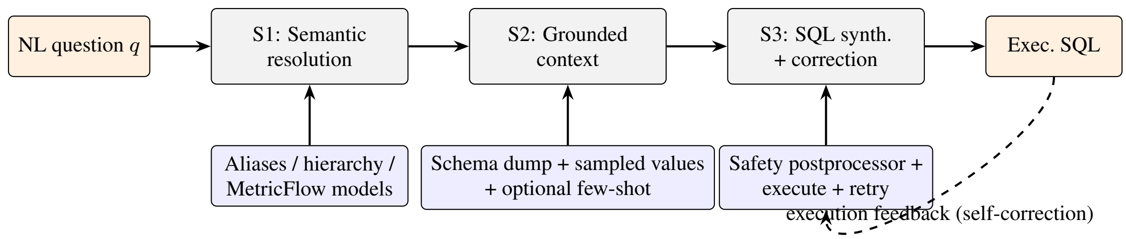

Figure 1: Architecture of NL2Semantic2SQL: Stage 1 semantic resolution, Stage 2 grounded-context assembly, Stage 3 SQL synthesis with postprocessor and self-correction.

3.4 Geospatial-aware grounding and bilingual routing

The geospatial path identifies geometry columns, geometry types, and spatial-reference information, then injects operator rules that distinguish topological from metric predicates and state when area or distance requires transformation or geography casting. When spatial signals are absent, the framework falls back to generic schema linking and aggregation-oriented prompts. A bilingual intent classifier recognises Chinese GIS terms, administrative names, and measurement units alongside English warehouse expressions, allowing one interface to serve GIS and warehouse questions without language-specific model swapping.

3.5 Warehouse-side enhancements

For non-geospatial warehouse queries, the framework adds three enhancements: candidate ranking that deprioritises geometry tables when no spatial signal is present, value-aware grounding from low-cardinality text columns, and MetricFlow-style fact/dimension metadata for join-path hints. These hints are injected only for non-spatial queries, leaving the GIS path unchanged.

3.6 Execution-time self-correction

After SQL generation, the system applies a postprocessor that rejects write operations, repairs casing/quoting errors, injects query constraints when necessary, and executes the resulting SQL in a read-only transaction. If execution fails, the framework enters a bounded self-correction loop. Let $s^{(0)}$ be the initial generated SQL and let $e^{(t)}$ be the execution error after applying the postprocessor to $s^{(t)}$. The retry policy is:

$$s^{(t+1)} = g\big(q, s^{(t)}, e^{(t)}, S\big), \quad t = 0, 1, \ldots, T-1, \tag{2}$$

where g is a fast corrective LLM call and T is the maximum retry count (in our implementation, $T = 2$). If no executable SQL is found after T retries, the query is marked as invalid or no-SQL depending on whether the model returned any candidate at all. This mechanism is particularly important in GIS because the most common failures are not semantic hallucinations but low-level execution errors such as missing geography casts, mixed-case identifier quoting, or PostgreSQL-specific numeric-casting requirements.

3.7 Worked example: end-to-end PostGIS grounding

A worked Chongqing example (length of one-way roads in km) is provided in Supplement S2, including the Stage 2 `::geography` cast, unit conversion, row-shape sanity check, and execution-based match. The companion repository releases minimal versions of the three implementation artefacts: the YAML semantic layer, the intent-rule set, and the safety postprocessor.

3.8 Multi-track evaluation protocol

Evaluation uses three tracks. The geospatial track is CQ-125 (85 Spatial + 40 Robustness). The warehouse track uses BIRD mini_dev (Li et al., 2023) with a 108-question design set and a 150-question held-out set. The cross-lingual track pairs 50 Chinese-translated BIRD questions with 50 native-Chinese GIS questions. Execution accuracy (EX) is the primary metric: predicted and gold result sets must match under set equality with numeric tolerance. Sample sizes, paired tests, and track construction are detailed in Section 4.

4 Experimental Setup

4.1 Experimental setup

Evaluation protocols. The primary analysis is the CQ-125 grounding-effect study in §6: no-grounding (A) versus grounding-only (B) across 11 LLM families, with N=3 paired samples and question-level majority-vote McNemar tests. A focused `gemini-3.5-flash` diagnostic adds grounding plus minimal-modification rules (C) and uses a balanced A/B/C design with N=5. Table 3 separates this inferential panel from supporting framework, deployment, and harness-readiness results.

Framework-level results in §5 use a separate protocol and provide context only. Early full-pipeline runs used an ADK agent loop that invoked grounding, execution, and retry tools over multiple turns, costing

Table 3: Evaluation-protocol inventory. MV denotes question-level majority vote before paired McNemar testing. A is no-grounding, B is grounding-only, and C is grounding plus the minimal-modification prompt profile.

Experiment	Dataset / model scope	Conditions and samples	Role in the manuscript
Cross-family grounding panel	CQ-125; 11 LLM families	A/B, N=3, MV McNemar	Primary evidence for subset-conditional and family-conditional grounding effects.
Focused Gemini 3.5 diagnostic	CQ-125; `gemini-3.5-flash`	A/B/C, balanced N=5, MV McNemar	Diagnostic check of the negative Spatial tail and mini-mod candidate patch.
Framework context	GIS, BIRD, cross-lingual, and DIN-SQL tracks	Original paired protocols; BIRD in single-pass mode	Supporting system context, not the primary grounding-effect claim.
Gemma4 fixed-host sweep	CQ-125; five local Gemma4 scales on host228	Single-host frozen summaries; best-observed full rows where multiple full runs existed	Deployment accuracy–runtime probe only.
FloodSQL-Bench smoke audit	Local PostGIS port; `gemini-2.5-flash`	Stratified 150q, N=1	External harness-readiness audit only.

$\sim$32$\times$ baseline tokens and sometimes entering unbounded tool-call cycles. We therefore added a deterministic *single-pass* pipeline: build grounding locally, make one SQL-generation LLM call, postprocess and execute, then retry with a corrective LLM up to $T = 2$ if execution fails. Single-pass reduced BIRD token cost to 7.9$\times$ baseline and improved design-set EX; BIRD results use this mode, while the earlier GIS framework headline remains separated from the cross-family grounding-effect analysis.

We also include a Gemma4 size sweep on the same local Ollama node (host228), covering `e2b`, `e4b`, `12b`, `26b`, and `31b`. It is a single-host engineering probe of accuracy–runtime trade-offs and is excluded from the N=3 majority-vote McNemar panel.

Benchmarks. We evaluate three tracks. The *GIS track* has 125 questions: 85 Spatial questions across three difficulty levels and 40 Robustness questions spanning security rejection, anti-illusion, schema hallucination, schema enforcement, data tampering prevention, and OOM prevention. The *BIRD track* uses BIRD mini_dev (Li et al., 2023) with a 108-question design set and a disjoint 150-question held-out set. The *cross-lingual track* has 50 Chinese-translated BIRD questions plus 50 native Chinese GIS questions.

For external-benchmark readiness, we ported FloodSQL-Bench (Liu et al., 2025a) to local PostgreSQL/PostGIS. The port contains 443 questions over 10 flood-risk, social-vulnerability, infrastructure, and boundary tables; 18 are reserved as few-shot examples. A stratified 150-question smoke run with `gemini-2.5-flash` produced identical baseline and full EX (65/150, 43.3%). We report it only as a harness-readiness audit, not as generalisation evidence.

Table 4: CQ-125 Spatial-subset profile. Predicate families may overlap because one question can require more than one spatial operation. The 40-question Robustness subset is reported separately.

Category	Count
Predicate family (a query may belong to several)	
topological	21
metric	20
knn	5
aggregation	46
plain	23
PostGIS function frequency	
`ST_Transform`	14
`ST_Intersects`	12
`ST_Area`	7
`ST_Length`	6
`ST_Contains`	6
`ST_Distance`	5
`<->`	5
`ST_AsText`	3
`ST_Within`	3
`ST_Centroid`	2
`ST_DWithin`	2
`ST_Union`	2
Multi-table join multiplicity	
single-table	55
2-table join	29
3+ table join	1
Difficulty stratification	
Easy	24
Medium	36
Hard	25
Total Spatial questions	85

EX is the primary metric: a prediction is correct only when its result set matches the gold SQL under set-equality with numeric tolerance. We also report Wilson intervals, execution validity, difficulty breakdowns, and paired McNemar tests. For repeated stochastic samples in §6, the question is the unit of inference: majority vote is taken per question and condition before one exact two-sided McNemar test. Repeated samples are not pooled as independent observations.

Execution accuracy evaluator. EX compares predicted and gold result sets with row-order invariance, positional column order, numeric tolerance (10^{-3} absolute or relative), NULL equality, three-decimal float rounding, gold-`LIMIT` handling, and duplicate-row preservation. The evaluator is a single Python function (`compare_results()`) in the companion repository; all 85 Spatial gold queries return at most 10 000 rows.

For the framework-level tracks, we compare two pipeline configurations:

- **Baseline**: direct LLM generation (Gemini 2.5 Flash) with schema dump only, no semantic grounding.
- **Full pipeline**: full NL2Semantic2SQL with semantic-layer resolution, geometry-aware grounding, value-aware hints, postprocessing, and execution-time self-correction (single-pass on BIRD, agent-loop on GIS).

We additionally compare against **DIN-SQL** (Pourreza and Rafiei, 2023) on framework-level tracks. These supporting configurations share the same database backend and evaluator. The cross-family grounding-effect study varies the LLM family while keeping A/B prompt conditions, benchmark, evaluator, and question-level test convention fixed. Full configurations are in Supplement S1.

5 Baseline Framework Evaluation

Scope note. This section gives compact framework-level context only. It is separate from the primary cross-family grounding-effect study in §6; detailed framework, BIRD, DIN-SQL, token-cost, and case-study tables are retained in Supplement S1. The $+32.50$ pp Robustness effect reported for `gemini-3.5-flash` in the cross-family table is the $N{=}3$ CQ-125 A/B panel value; the focused balanced $N{=}5$ A/B/C diagnostic reports $+27.50$ pp for the same family and subset. Neither number is the earlier 15-question pilot noted below.

5.1 Supporting framework results

In the earlier agent-loop GIS run, the best observed framework sample reaches Spatial EX 0.706 on the 85-question set with a `gemini-2.5-flash` backbone, versus a schema-only baseline of 0.518. This best-sample result is descriptive and is not used as primary inferential evidence. On the 15-question Robustness pilot, the framework reaches $7/7$ on discordant pairs ($p = 0.0156$ exact). On BIRD held-out 150q, the Full pipeline reaches EX 0.507 vs. DIN-SQL 0.440 ($p = 0.0755$, marginal). On the 40-question Robustness suite, the gap concentrates on safety/refusal categories where DIN-SQL has no built-in handling (Full 0.975 vs. DIN-SQL 0.275). These supporting results motivate the framework context; the main CQ-125 claims use the question-level majority-vote protocol in §6.

5.2 Gemma4 model-size sweep on a fixed local host

As a deployment-oriented probe, we ran five Gemma4 model scales on one fixed local Ollama serving node using CQ-125 and the same baseline-vs-full framework configurations. The source table is provided as `gemma4_host228_scale_sweep_summary.csv`. The serving metadata were queried from the same node via the Ollama `/api/tags` and `/api/version` endpoints: Ollama version 0.30.5; GGUF format; Q4_K_M quantisation for all five Gemma4 models; reported parameter sizes of 5.1B, 8.0B, 11.9B, 25.8B, and 31.3B. The semantic retrieval backend on the same node used `nomic-embed-text-v2-moe`. The full model-metadata source table is distributed as `host228_ollama_model_metadata.csv`. This probe is descriptive: it is not an inferential cross-family test and is not used in the primary majority-vote McNemar reductions.

In this fixed-host summary, full-pipeline EX ranged from 80/125 to 114/125, compared with 37/125 to 71/125 for schema-only generation. The `Gemma4:31b` full run had the highest observed EX (114/125, 20.04 min), while `Gemma4:26b` was one question lower (113/125, 12.50 min). These numbers are reported only to illustrate an accuracy–runtime deployment trade-off. Because several host228 full-pipeline runs existed for some scales, the CSV identifies the selected full rows as best-observed engineering summaries rather than stochastic significance evidence; the selection procedure precludes inferential comparison across model sizes.

5.3 Reproducibility

All commands and artefacts to regenerate the previously released framework-level reductions are in the public GitHub reproducibility repository. The reviewer runtime is $\approx 90\,\mathrm{s}$ with zero LLM API and zero PostgreSQL dependency. These checks reproduce tables from frozen SQL-output and execution-result records rather than re-executing queries against the withheld operational database rows. The cross-family analysis in §6 is regenerated from the same repository. The newer Gemma4 deployment probe is reported from frozen local run summaries; its source-table summary is included with this manuscript source as `gemma4_host228_scale_sweep_summary.csv` and is not used in the primary statistical reductions.

6 Subset-Decomposed Grounding Effects in NL2GeoSQL

6.1 Phenomenon: grounding effects depend on subset and model family

Schema-aware grounding is expected to reduce hallucination and improve schema linking, but in PostGIS NL2GeoSQL it also affects predicate choice, geometry measurement, joins, output shape, refusal, and bounded execution. We therefore evaluate grounding by subset rather than aggregate accuracy alone.

The primary panel is the 11-family $B - A$ comparison in Table 7. Under question-level majority vote across $N = 3$ samples, grounding improves Robustness for all 11 families by point estimate. Spatial effects are heterogeneous: most families improve, but `gemini-3.5-flash` has the most negative panel estimate (-10.59 pp, $p = 0.136$) and `gemini-3.1-pro-preview` is near zero negative (-2.35 pp, $p = 0.845$). This establishes heterogeneity, not a significant family-level regression.

A focused three-condition `gemini-3.5-flash` analysis, balanced at $N = 5$ for no-grounding (A), grounding-only (B), and grounding plus minimal-modification (C), gives the same diagnostic signal. A versus B changes Spatial by -10.59 pp ($b/c = 19/10$, $p = 0.136$) and Robustness by $+27.50$ pp ($p = 0.0010$); the aggregate change is only $+1.60$ pp ($p = 0.875$). We therefore treat the negative Spatial tail as diagnostic rather than confirmatory.

Statistical convention. Unless otherwise stated, all paired tests in this section use **question-level majority vote across the N stochastic samples followed by a single McNemar exact two-sided test** on the resulting per-question table. This avoids the pseudoreplication that would arise from pooling repeated stochastic observations of the same question. The convention is applied uniformly to the three-condition analysis and to the cross-family panel.

6.2 Three-factor decomposition

We decompose the gain space using three conditions per question, paired by sample index:

- **A: no-grounding** — bare schema dump, no semantic-layer hints. (N=5.)
- **B: grounding only** — standard grounding context (alias hints, value-enum, large-table flags), no profile patch. (N=5.)

- **C: grounding + minimal-modification (mini-mod)** — grounding plus seven *no unsolicited X* rules targeting over-engineering. (N=5.)

Subset	A: no-grounding	B: grounding only	C: grounding + mini-mod
Overall (125q)	53.6%	55.2%	**60.8%**
Robustness (40q)	65.0%	**92.5%**	**92.5%**
Spatial (85q)	**48.2%**	37.6%	45.9%
Easy (24q)	**66.7%**	54.2%	**66.7%**
Medium (36q)	**52.8%**	38.9%	44.4%
Hard (25q)	24.0%	20.0%	**28.0%**

Table 5: Question-level majority-vote execution accuracy across N=5 samples for `gemini-3.5-flash` on CQ-125 under three conditions. Bold marks the highest cell per subset. The headline observation is that on the Spatial subset, the no-grounding condition (A: 48.2%) outperforms both grounding-only (B: 37.6%) and grounding+mini-mod (C: 45.9%); on the Robustness subset the order reverses sharply (A: 65.0% < B/C: 92.5%). The aggregate at C is therefore a weighted compromise, not a uniformly Pareto-better operating point.

Paired Δ decomposition. Table 6 reports paired Δ per cell under the question-level majority-vote McNemar convention. The grounding effect ($B - A$) is large and significant on Robustness (+27.50 pp, p=0.0010), and it is negative but non-significant on Spatial in this focused analysis (−10.59 pp, p=0.136). The mini-mod patch effect ($C - B$) recovers +8.24 pp on Spatial but does not reach α=0.05 (p=0.065, marginal). The total mini-mod-against-baseline effect ($C - A$) is non-significant on Spatial (−2.35 pp, p=0.84). **Mini-mod therefore remains a candidate partial mitigation, not a statistically validated recovery.**

Subset	Grounding ($B - A$)	Mini-mod patch ($C - B$)	Total ($C - A$)
Overall (125q)	+1.60 pp (p=0.87)	+5.60 pp (p=0.092)	+7.20 pp (p=0.19)
Robustness (40q)	**+27.50** pp (p=0.0010)	+0.00 pp (p=1.00)	**+27.50** pp (p=0.0034)
Spatial (85q)	−10.59 pp (p=0.136, n.s.)	+8.24 pp (p=0.065, marginal)	−2.35 pp (p=0.84, n.s.)
Easy (24q)	−12.50 pp (p=0.51)	+12.50 pp (p=0.38)	+0.00 pp (p=1.00)
Medium (36q)	−13.89 pp (p=0.27)	+5.56 pp (p=0.63)	−8.33 pp (p=0.55)
Hard (25q)	−4.00 pp (p=1.00)	+8.00 pp (p=0.50)	+4.00 pp (p=1.00)

Table 6: Paired Δ ($\pm$ McNemar exact two-sided p) under question-level majority-vote across N=5 samples. Bold marks results significant at p<0.05. The Spatial grounding effect in this focused `gemini-3.5-flash` analysis remains negative ($B - A = -10.59$ pp), but it is non-significant. The mini-mod patch effect on Spatial ($C - B = +8.24$ pp) is marginal; the total $C - A$ is non-significant.

Why de-pooled tests matter here. A naive pooled test would treat 85 Spatial questions $\times$ N=5 samples as 425 independent paired observations, drastically inflating the effective sample size. Same-question samples are not independent: they share question difficulty, schema-mapping ambiguity, and gold-SQL idiosyncrasies. The honest design is one paired observation per question (the within-sample majority vote), giving 85 Spatial observations. The price is statistical power; the gain is a defensible p-value.

Mixture sensitivity. The focused `gemini-3.5-flash` A/B contrast shows why aggregate EX depends on benchmark composition. Let w be the Spatial share of a mixed benchmark. The expected aggregate grounding effect is

$$\Delta(w) = w(-10.59) + (1 - w)(27.50).$$

The effect changes sign at $w \approx 0.722$. CQ-125 has $w = 85/125 = 0.68$, so the aggregate is close to zero despite opposite-signed subset effects. This analytic check adds no data; it makes composition sensitivity explicit.

6.3 Cross-family evidence: heterogeneous grounding effects across 11 LLMs

We extend the analysis to ten additional LLMs whose CQ-125 N=3 baseline-vs-full data are available from prior batch runs. None uses mini-mod; we report only $B - A$ paired Δ under the same question-level majority-vote McNemar convention. This is the primary cross-family evidence in the paper.

	Spatial (85q)			Robust (40q)		
Family	A mean %	B mean %	MV Δ (pp)	A mean %	B mean %	MV Δ (pp)
`gemini-2.5-flash`	54.9	68.6	$+12.94^{*}$	40.0	97.5	$+57.50^{**}$
`gemini-2.5-pro`	57.6	72.5	$+18.82^{**}$	61.7	91.7	$+30.00^{**}$
`gemini-3.1-flash-lite-preview`	50.2	67.8	$+16.47^{**}$	70.0	85.0	$+15.00$ (n.s.)
`gemini-3.1-pro-preview`	65.1	63.5	-2.35 (n.s.)	63.3	99.2	$+35.00^{**}$
`gemini-3.5-flash`	48.6	36.5	$\mathbf{-10.59}$ *(p=0.14)*	58.3	89.2	$+32.50^{**}$
`deepseek-v4-flash`	43.1	62.8	$+23.53^{**}$	28.3	81.7	$+62.50^{**}$
`deepseek-v4-pro`	48.2	61.6	$+15.29$ (p=0.05)	28.3	90.0	$+65.00^{**}$
`qwen3.6-flash`	35.3	49.8	$+12.94$ (p=0.06)	45.8	72.5	$+27.50^{**}$
`qwen3.6-plus`	50.2	65.1	$+12.94^{*}$	66.7	91.7	$+30.00^{**}$
`qwen3.7-max`	44.3	64.3	$+17.65^{**}$	73.3	94.2	$+22.50^{*}$
`gemma-4-31b-it`	44.7	57.3	$+11.76^{*}$	89.2	95.8	$+12.50$ (p=0.06)

Table 7: Cross-family question-level majority-vote Δ on CQ-125 (N=3 samples per family). A and B columns report per-sample mean execution accuracy across the three samples for readability; MV Δ is computed after question-level majority vote as MV(B) − MV(A) in percentage points. Significance is the paired exact two-sided McNemar test on the per-question majority-vote table. Significance markers are unadjusted per-cell values: $^{*}p<0.05$, $^{**}p<0.01$. Bold-italic marks the most negative panel point estimate (`gemini-3.5-flash`, p=0.136). Of 11 families, 7 show unadjusted significant positive Spatial Δ, 2 marginal positive (p between 0.05 and 0.10), and 2 non-significant negative. Robustness Δ is positive for every family, with nine unadjusted significant, one marginal, and one non-significant positive cell.

Multiplicity. The star markers in Table 7 are unadjusted per-cell p-values retained to show the panel pattern. Treating the 11 Spatial and 11 Robustness tests as two exploratory comparison sets, Holm correction at α=0.05 leaves 3 Spatial cells and 9 Robustness cells significant; Benjamini–Hochberg FDR control at q=0.05 leaves 6 Spatial cells and 9 Robustness cells significant. We therefore read counts of significant family cells descriptively. The confirmatory claim is the subset-conditional heterogeneity pattern under the de-pooled question-level convention, not a corrected discovery claim for every individual family.

What the table establishes and what it does not. Grounding produces a *heterogeneous* family-conditional Spatial effect, ranging from -10.59 pp (`gemini-3.5-flash`) to $+23.53$ pp (`deepseek-v4-flash`). Robustness point estimates are positive for all 11 families ($+12.50$ to $+65.00$ pp). The table does *not* establish a stable negative property of `gemini-3.5-flash`: its Spatial estimate is the most negative but non-significant, and the balanced $N = 5$ check supports diagnosis rather than confirmation.

Aggregate metrics are insufficient. Aggregate execution accuracy on a mixed Robustness–Spatial benchmark is a poor proxy for the interaction: a small positive Overall Δ can be driven by Robustness gains while Spatial gains vary by family. Subset decomposition with question-level paired tests is therefore necessary for this benchmark type.

6.4 Mini-mod profile and its (limited) effect

The mini-mod profile prepends seven *no unsolicited X* rules to the system prompt of `gemini-3.5-flash` only:

1. No unsolicited `GROUP BY` unless explicit grouping language is present;
2. No unsolicited aggregation (`SUM`/`AVG`/`MAX`/`MIN`/`COUNT`);
3. No unsolicited `ST_Union`;
4. No unsolicited `ROUND`;
5. No unsolicited nested subqueries; flat `SELECT` preferred;
6. No unsolicited `ST_Transform`;
7. Treat grounding hints as reference, not checklist.

These rules target the dominant A-pass/B-fail patterns for `gemini-3.5-flash`: nested subqueries (29.7%), redundant `WHERE` clauses (17.2%), unsolicited `GROUP BY` (14.1%), and `ST_Union` pre-aggregation (4.7%).

Empirical effect under de-pooled statistics. Per Table 6, mini-mod does *not* reach α=0.05. Its Spatial $C-B$ point estimate is marginal (+8.24 pp, p=0.065), and $C-A$ remains non-significant (-2.35 pp, p=0.84). We therefore treat mini-mod as **a candidate prompt-level partial mitigation, not a validated production technique**.

6.5 Design implication: intent-aware grounding gating as upper-bound analysis

Table 5 suggests a design direction, not a validated production policy. On `gemini-3.5-flash`, Robustness queries gain from grounding (B/C, 92.5%), while Spatial queries perform best under no-grounding (A, 48.2%). A hypothetical perfect intent classifier would yield an upper-bound EX of $\frac{40\cdot 92.5+85\cdot 48.2}{125} \approx 62.4\%$, above any single static policy (A=53.6%, B=55.2%, C=60.8%).

What we report and what we do not. This is an *upper-bound theoretical analysis*. We do not report end-to-end EX under a real classifier. A complete gating evaluation must report classifier P/R/F1, EX after misrouting, and latency/token overhead. The contribution here is the upper-bound number and the qualitative implication that subset-conditional grounding is a plausible policy class for this family.

For all other tested families, plain grounding remains the empirically best static policy at N=3; the gating logic is family-conditional.

6.6 Methodological lesson

We propose a compact analysis template for mixed Spatial–Robustness grounding studies:

1. **Data manifest first.** Enumerate every $N\geq 3$ JSONL before constructing tables.
2. Run three conditions per family (A no-grounding, B grounding-only, C grounding+patch) at $N\geq 5$ where feasible.
3. Report Overall, Robustness, Spatial, Easy/Medium/Hard, and category breakdowns where $n_q\geq 3$.

4. **Per subset, use a question-level paired test.** Majority-vote repeated samples per question, then run one exact two-sided McNemar test; do not pool repeated samples as independent observations.
5. Report decomposition $B - A$, $C - B$, $C - A$ separately; any narrative must attribute the gain to one of these.
6. State the multiplicity policy before counting significant family cells; if correction is exploratory rather than confirmatory, say so explicitly.
7. Pre-specify competing interpretations, including null and negative outcomes.

Every number in Tables 5–7 is regenerable from the released JSONL via the offline analysis scripts shipped in the companion repository.

6.7 Section-level limitations

The empirical section has four boundaries carried into the Discussion: the cross-family panel is $N=3$ while only the focused diagnostic is $N=5$; CQ-125 is one Chongqing-grounded benchmark with an 85:40 Spatial–Robustness mixture; mini-mod is a candidate patch; and intent-aware gating is reported only as an upper bound, not as classifier-validated end-to-end EX.

6.8 Reproducibility

All numerical results in this section — the three-factor decomposition (Tables 5–6) and the cross-family Table 7 — are regenerable offline from frozen evaluation records with zero LLM API calls, zero PostgreSQL dependency, and zero GPU requirement. This supports reproducibility of the statistical reductions from frozen SQL-output and execution-result records; it is not a raw-row re-execution package because the operational database rows are withheld. The public GitHub reproduction repository is available at `https://github.com/zhouning/nl2geosql-reproduction`. It ships frozen JSONL execution traces, benchmark question metadata, prompt profiles, and offline analysis scripts for all eleven LLM families. Each primary panel run record is keyed by `(family, sample_index, mode, qid)` and can be reduced by the question-level majority-vote convention used in this manuscript. The package also releases the schema-level materials needed to audit grounding without exposing sensitive row-level records: a structure-only `schema.sql` dump, table/column metadata, geometry-column metadata, SRID information, semantic-layer configuration, the minimal-modification prompt profile, and the upper-bound intent-gate decision template. Embeddings used during semantic resolution are precomputed and cached in the JSONL traces, so reviewers do not need to invoke any embedding model. The repository includes a CI-style `verify_tables.py` check for one frozen cross-family table; the majority-vote reductions reported here are computed from the same JSONL records under the de-pooled convention stated above.

7 Discussion

7.1 A Spatial–Robustness inversion in one fast-model family

Per-record diff between A (no grounding) and B (grounding only) on `gemini-3.5-flash` suggests that many regressions involve SQL elaboration the question did not ask for. Among 91 distinct A-pass / B-fail pairs across $N = 3$ samples (pooled for diagnostic, not for inference), the four largest patterns are nested subquery wrapping (29.7%), redundant `WHERE` clauses (17.2%), unsolicited `GROUP BY` (14.1%), and `ST_Union` pre-aggregation before `ST_Centroid`/`ST_Area` (4.7%). Each can shift the result-set shape away from the gold answer even when the model has identified the relevant table and spatial concept.

A geospatial cognitive-load reading. One GIScience interpretation is that grounding context exposes more table-and-column metadata than the question uses, encouraging `gemini-3.5-flash` to write "explanatory" SQL that aggregates across parent tables, joins to context tables, or projects additional columns. We call this diagnostic pattern *geospatial over-engineering*: spatial evidence such as geometry types, SRIDs, predicate taxonomies, and alias hierarchies may become checklist material rather than reference material for some model families. Spatial questions are more exposed because they admit many plausible-but-wrong elaborations; Robustness questions are constrained by refusal, bounded preview, and schema validation. Baseline headroom may also matter: families with weaker no-grounding Spatial EX, such as `deepseek-v4-flash`, `qwen3.6-flash`, and `qwen3.7-max`, may gain more from added schema information. We do not have access to model internals, so this remains an interpretation rather than a mechanistic claim.

7.2 Why subset decomposition is not optional

The CQ-125 benchmark mixes 85 spatial-query questions and 40 robustness questions. For the same family, the focused three-condition analysis gives opposite-signed subset effects for grounding-only versus no-grounding: -10.59 pp on Spatial and $+27.50$ pp on Robustness. Any aggregate metric over a benchmark with this kind of mixture can obscure the interaction. This lesson is directly relevant to GIScience evaluation: spatial predicate translation, metric measurement, bounded preview, and refusal are different behaviours and should not be collapsed into a single headline without subset reporting.

7.3 Model-size scaling is a deployment question, not only a leaderboard question

The fixed-host Gemma4 sweep is a deployment probe. Under the same CQ-125 benchmark and local Ollama host, the full pipeline improved all five tested Gemma4 scales by descriptive point estimate. `Gemma4:31b` had the highest observed full-pipeline EX (114/125, 20.04 min), while `Gemma4:26b` was one question lower (113/125) and faster (12.50 min). These single-host summaries are not family-level statistical evidence; they show that deployment choice can involve an accuracy–runtime trade-off rather than a largest-model rule.

7.4 Limitations

1. **Finite-sample regime; multiple-comparison correction.** All primary cross-family panel cells are at $N{=}3$ paired samples per question. The focused `gemini-3.5-flash` A/B/C check is balanced at $N{=}5$ and retains a negative Spatial point estimate, but it remains non-significant ($p = 0.136$). Cross-family table markers are unadjusted per-cell p-values used descriptively. Holm correction within the 11 Spatial family tests leaves three positive cells significant and does not make the `gemini-3.5-flash` negative tail significant; Holm correction within the 11 Robustness tests leaves nine positive cells significant. We label the negative tail a diagnostic observation, not a confirmed model-family property.

2. **Single primary benchmark.** CQ-125 has a 85:40 spatial:robustness mixture. We have ported FloodSQL-Bench to the same PostgreSQL/PostGIS evaluation harness and completed a single `gemini-2.5-flash` stratified smoke run, but that run is not a full external replication. Confirmation requires a pre-specified multi-family, $N{\geq}3$ run on benchmarks of different composition,

including pure-spatial GeoSQL-Eval (Hou et al., 2026), FloodSQL-Bench (Liu et al., 2025a), and a pure-warehouse BIRD held-out set.

3. **Mini-mod is a candidate patch.** The balanced $N{=}5$ patch effect on Spatial is marginal ($p = 0.065$) and we do not claim it as a validated technique.

4. **Intent-aware gating is upper-bound analysis.** A classifier-validated end-to-end EX evaluation, including classifier P/R/F1 on CQ-125 and end-to-end EX after misrouting, is not reported in this manuscript and is identified as immediate follow-up.

5. **Subset of LLMs surveyed.** The 11-family panel includes only models with API access during the evaluation window. Replication on additional model providers is required before treating the observed distribution of effects as representative of the broader model ecosystem.

6. **Gemma4 scale sweep is a deployment probe.** The five-size Gemma4 sweep is reported to expose a practical accuracy–runtime trade-off on a fixed local host. It is not a stochastic $N{=}3$ family-level inference panel, and its best-observed full-pipeline rows should not be interpreted as statistical significance evidence.

7. **Question phrasing.** Four of 125 questions admit superset-column readings. They are two Easy items and two Medium items in CQ-125. A relaxed col-count ablation does not change the qualitative conclusion.

7.5 Future work

(i) External replication of the balanced A/B/C diagnostic on benchmarks with different composition, including a pure-spatial PostGIS benchmark, FloodSQL-Bench, and a pure-warehouse benchmark. (ii) End-to-end intent-aware-gating evaluation under a real classifier with reported P/R/F1 and post-misrouting EX. (iii) Extension of the $N{\geq}5$ three-condition protocol beyond `gemini-3.5-flash` to additional LLM families. (iv) Inclusion of OpenAI and Anthropic families to test whether the family-conditional pattern is specific to the currently available model panel. (v) Mechanistic study of the over-engineering interpretation through controlled perturbation of grounding-context length and content.

8 Conclusion

We report a question-level paired-McNemar evaluation of schema-aware grounding across 11 LLM families on the 125-question PostGIS NL2GeoSQL benchmark CQ-125. The grounding effect is heterogeneous and family-conditional. On the 85-question Spatial subset, point estimates range from -10.59 pp (`gemini-3.5-flash`, $p = 0.136$) to $+23.53$ pp (`deepseek-v4-flash`, $p < 0.001$). On the 40-question Robustness subset, all 11 families improve in point estimate. A balanced $N{=}5$ three-condition analysis of `gemini-3.5-flash` retains the negative Spatial point estimate for grounding-only versus no-grounding (-10.59 pp, $p = 0.136$), but does not convert it into a significant family-level regression. The methodological conclusion is stronger: aggregate execution accuracy is insufficient on mixed Spatial–Robustness benchmarks, and pooled McNemar tests over repeated stochastic samples inflate significance. We recommend subset-decomposed reporting and question-level majority-vote McNemar tests as the default evaluation protocol for grounding studies in geospatial NL2SQL.

Author Contributions

Ning Zhou conceived the study, designed the benchmark and semantic-layer evaluation protocol, implemented the analysis scripts, ran the experiments, verified the statistical analyses, wrote the manuscript, and approved the final version for submission.

Funding

The author received no financial support for the research, authorship, or publication of this article.

Competing Interests

The author declares no competing interests.

Ethics Statement

This study uses benchmark questions, schema-level database metadata, generated SQL, and offline execution/evaluation records. It does not report experiments on human participants, patient data, or animal subjects. The underlying operational database rows are withheld because they contain sensitive local data.

Use of Artificial Intelligence Tools

Large language models are the objects of evaluation in this study, and their use in the experimental protocol is described in the manuscript. Claude Code and OpenAI Codex were also used for editorial support, language polishing, format checking, and consistency review during manuscript preparation. The author checked and revised all AI-assisted text, verified the accuracy of the manuscript and references, and takes full responsibility for the final content. No AI tool is listed as an author.

Data Availability

The benchmark questions, schema-level metadata, prompt profiles, frozen LLM-output records, and execution-result records needed to audit the manuscript's reported reductions are available from the public GitHub reproduction repository: `https://github.com/zhouning/nl2geosql-reproduction`. The released materials support reduction-level reproducibility from frozen records. They do not require LLM API calls, database access, GPU resources, or access to sensitive raw records. The underlying operational database rows are not released because they contain sensitive local data. Instead, the repository provides the schema-level information needed for auditability, including table/column structure, geometry-column metadata, SRID information, and semantic grounding rules. The balanced `gemini-3.5-flash` N=5 A/B/C diagnostic summary, the fixed-host Gemma4 deployment sweep, and the FloodSQL-Bench smoke audit added in this revision are provided as source-table summaries with the manuscript source and are not used as replacements for the primary statistical reductions.

Computer Code Availability

The computer code used to produce the offline reductions reported in this manuscript is open-source and freely available for download from the public GitHub reproduction repository: `https://github.com/zhouning/nl2geosql-reproduction`. The repository is publicly accessible and does not require credentials for download or inspection.

The repository contains individual source files rather than a compressed code archive. It includes a `README` file describing the purpose of the code, installation requirements, the repository layout, and step-by-step reproduction commands. It also includes quick-test/example checks for the main manuscript reductions. In particular, the commands `python tables/build_codex_tables.py` and `python tables/verify_codex_tables.py` rebuild and verify the question-level majority-vote and McNemar-test results used in the revised manuscript. The checks run on frozen JSONL execution records and do not require proprietary software, LLM API calls, PostgreSQL/PostGIS access, GPU resources, or access to the withheld operational row-level database. The repository license file specifies MIT licensing for code and CC-BY 4.0 licensing for benchmark questions and frozen LLM evaluation outputs under `data/`.

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